

Biomagnification of cycad neurotoxins in flying foxes: implications for ALS-PDC in Guam.

Beta-methylamino-L-alanine (BMAA) occurs in higher levels in museum specimens of the Guamanian flying fox than in the cycad seeds the flying foxes feed on, confirming the hypothesis that cycad neurotoxins are biomagnified within the Guam ecosystem. Consumption of a single flying fox may have resulted in an equivalent BMAA dose obtained from eating 174 to 1,014 kg of processed cycad flour. Traditional feasting on flying foxes may be related to the prevalence of neuropathologic disease in Guam.